Comparative Analysis of Machine Learning Models for Diabetes Prediction

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Abstract**—**Diabetes remains one of the most pressing global public health issues, where early detection can facilitate prompt interventions and prevent diabetes related complications. This study explores the potential of machine learning methods in diabetes prediction on the CDC BRFSS 2015 Diabetes Health Indicators dataset. This dataset is highly imbalanced with a very small number of cases with diabetes as compared to the large number of non-diabetic cases, and that makes the diabetes classification task more difficult. In order to solve this problem first an anomaly detection method: Isolation Forest is compared with several traditional supervised machine learning models, including Logistic Regression, Naive Bayes, Decision Tree, and Random Forest. The models are compared using accuracy, precision recall, F1-score, and AUC-ROC metrics. More importance is given to recall and F1-score, as these measures provide more useful information for identifying diabetic cases in an imbalanced dataset than accuracy, which is visually implemented. The findings reveal that the performance is greatly influenced by model selection and also indicate the compromise between overall accuracy and the capacity to identify positive diabetes cases. This work is focused on finding out if anomaly detection can be an alternative to conventional classifiers and to getting a good model for diabetes prediction using structured healthcare data.

Keywords—diabetes prediction, machine learning, anomaly detection, isolation forest, classification, imbalanced dataset, healthcare analytics

GitHub Link:
https://github.com/kiran-bhandari16/machine-learning-cs-7830-01-final-project

I. Introduction

Diabetes is one of the most widespread chronic diseases worldwide and if remains untreated, can cause complicated health issues. It is very important to be able to predict diabetes risk at an early stage so that patients can get the right treatment and also change their lifestyles before it is too late. Due to the increase in medical data, machine learning has become an excellent tool for finding patterns that might help in predicting the risk of disease.We use the CDC Diabetes Health Indicators dataset to carry out a study on diabetes prediction in this project.

This dataset contains almost all the health-related features based on which it is possible to determine the likelihood of a person being diabetic or not. The problem with this dataset is that it is highly imbalanced, with non-diabetic records being the majority and diabetic ones the minority. Hence a model with a high level of accuracy might still be very poor when it comes to identifying diabetic individuals. Thus not only is the problem more challenging but it is also more meaningful when it comes to applications in healthcare."Most of the previous studies that have been done on diabetes prediction have primarily concentrated on traditional supervised learning methods such as Logistic Regression, Decision Tree, Random Forest, and Naive Bayes.

Because of their efficiency, and simplicity in explanation, these models are commonly used. But in cases where the data is imbalanced, one should consider using anomaly detection methods, since diabetic cases can be seen as the rarer ones in the dataset. Following this logic, the present project also explores and isolates the Forest to compare it with the other traditional machine learning models.

The primary objective of this study is to contrast the use of anomaly detection and conventional machine learning methods for diabetes prediction. Models are built and tested on the CDC Diabetes Health Indicators dataset, and their results are measured by accuracy, precision, recall, F1-score, and AUC-ROC.

Because it is crucial to correctly identify diabetic cases, recall and F1-score are given more focus. This paper makes the following contributions:

- We explore the performance of traditional machine learning models against an anomaly detection technique for diabetes prediction on a highly skewed dataset.
- We assess the models by several performance metrics that are appropriate for imbalanced classification.
- We determine which method is better at detecting the most probable diabetic cases from healthcare data.

II. Related Work

A. Supervised Learning Approaches

A lot of the earlier research has used machine learning methods to analyze diabetes prediction. Supervised models are very popular, including Logistic Regression, Decision Tree, Random Forest, Naive Bayes, and neural networks. These methods are able to identify the patterns in clinical or health-indicator data and make a guess as to whether the patient might be diabetic or not. To mention a few examples, Rajendra et al. used logistic regression and machine learning methods to predict diabetes on the basis of diagnostic measurements. Rastogi and Andersen, on the other hand, compared various methods such as Random Forest, Support Vector Machine, Logistic Regression, and Naive Bayes for diabetes-related classification tasks.

Such studies support the usefulness of traditional supervised classifiers for structured health data.One of the most common observations that came out of this literature is that supervised models very often score quite well overall however depending on the dataset and distribution of the classes, their results can be quite different. Most of the diabetes datasets, for instance, contain a larger share of non-diabetic cases than diabetic ones. This is a typical example of a strong accuracy but weaker recall for the positive class. Our project is working along these paths by comparing several typical supervised models, such as Logistic Regression, Naive Bayes, Decision Tree, and Random Forest, on a binary diabetes prediction problem. [1] [2]

B. Self-Supervised and Unsupervised Methods

Alongside the traditional supervised learning, several research have examined anomaly-based or unsupervised approaches to healthcare forecasting, particularly in situations where the positive class is scarce. Isolation Forest is a very popular anomaly detection algorithm that identifies unusual data points by using randomly generated segments of data. Since minority or abnormal cases could look

somewhat different from the major class, anomaly detection techniques have been regarded as a possible means to tackling the imbalance of class problems. In fact, the scikit-learn documentation also describes Isolation Forest as an unsupervised outlier detection technique produced to single out the observations that can be isolated easier than the normal ones.On the other hand, anomaly-based methods come with some drawbacks for disease forecasting as well.

In contrast to supervised classifiers, they don't use the actual disease label to effectively guide training. Consequently, they might find unusual patterns but lacking the capability to generate the best classification result for a labeled diabetes task. Therefore, in our work, we use Isolation Forest primarily as a benchmark for comparison to see if anomaly-style modeling can be a good way to come up with diabetic cases in an imbalanced context as opposed to the traditional supervised classifiers. [3] [4]

C. Foundation Models and Transfer Learning

Very recently, the healthcare machine learning scene has shifted its focus towards foundation models and transfer learning, particularly when dealing with large electronic health record (EHR) datasets. Transformer-based pretraining adaptations to structured medical records followed by model fine-tuning for downstream disease prediction tasks are the main features of models like BEHRT and Med-BERT. Transformers for EHRs allowing the prediction of different future conditions was the main characteristic of BEHRT.

Med-BERT demonstrated that with a large-scale structured EHR pretraining, disease prediction downstream could be enhanced, especially when labeled data are scarce.Despite the fact that these methods bring hope, they are predominantly tailored for extensive longitudinal EHRs and are dependent on significant computational power, pretraining data, and sophisticated modeling pipelines. Our work takes a different path as it is geared to a structured tabular public dataset and exploits simple, interpretable, and readily available machine learning techniques that are less demanding in terms of implementation and analysis, which suits the course project context very well. Unless going for large pretrained clinical models, we are also using the opportunity to explore the performance of traditional supervised classifiers and an anomaly detection approach for diabetes prediction. [5] [6]

III. Methodology

A. Problem Formulation

The goal of this project is to use health-related data to guess if a person has diabetes or not. A vector $x \in R^d$, where d is the number of features of the input, such as BMI, age, general health, high blood pressure, and other health indicators, can be used to make a prediction about diabetes. The class variable is y where y=1 means the individual suffers from diabetes and y=0 means that the individual does not have diabetes. A machine learning model was created, which is $f\theta(x)$ where θ represents parameters that are learned during training.

Given input features, the model outputs a prediction for the target class. The training goal is to find the best model parameters by minimizing the loss function:

$$L(\theta) = \Sigma\, \ell(f\theta(x_i), y_i) + \lambda\, R(\theta) \qquad (1)$$

with ℓ being a function that quantifies how far the prediction deviates from the actual label, R(θ) is the regularization term which is used to control the complexity of the model, and λ is the parameter that sets how strictly the regularization is enforced.

Here, we divided the problem into a binary classification one, as there are only two classes in the output: diabetic and non-diabetic. Several machine learning algorithms were trained and the performances were compared so as to find out which model is the best one in terms of prediction on this particular dataset.

B. Model Architecture

This project simply does not use a deep learning architecture with multiple hidden layers. Rather, it works by using a number of traditional machine learning classification models to predict diabetes status of a person based on health-related features.

The models used in this paper are Logistic Regression, Naive Bayes, Decision Tree, Random Forest, and Isolation Forest.It turns out that the Random Forest model overall achieved the highest performance stats and this was the one we decided to go with as the final model. Random Forest belongs to ensemble learning techniques which combine many decision trees. Each tree is trained on a different subset of the original data that was randomly sampled and also on a subset of input features that was randomly sampled. The final decision is reached by taking the median of the decisions made by individual trees.

That way prediction results become both less prone to overfitting and more robust.The final Random Forest model was working with the following best performing hyperparameters: 100 trees, maximum depth of 20, minimum samples split of 8, minimum samples leaf of 4, and square root feature selection at each split. Thanks to these parameter setups, the model was able to capture the significant patterns in the health indicators together with its features, while at the same time it was able to keep the complexity of the model within reasonable bounds.On the contrary to neural networks, this model is built without hidden layers or activation functions.

In fact, it is an ensemble composed of multiple decision trees. To further understand the model, a feature importance analysis was carried out to establish which input features were the biggest contributors to the output of the model. The top features have been found to be BMI, general health, high blood pressure, and age.

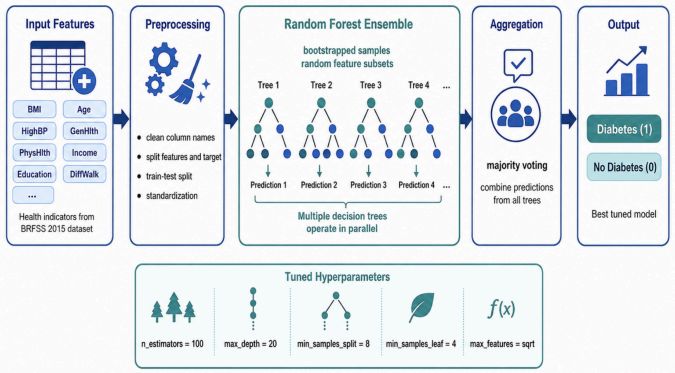

Fig. 1. Architecture of Proposed Random forest model for diabetes prediction

C. Training Procedure

The project was developed using Python programming language with machine learning libraries like scikit-learn pandas NumPy, and Matplotlib. Since the work focuses on traditional machine learning models, the training procedure differentiates from that of neural network training where the use of optimizers such as Adam or SGD, scheduling of learning rate and epoch-based training process are common.To begin with, the data set was cleaned and made ready for modeling.

Changes were made to the column names so they conformed with standards, input features and target variables were separated, the data was split into training and testing sets. Model performance evaluation on unseen data was done through a standard train-test split. Feature

scaling was also done on the training set to make sure if the models were working rightly and it was fair especially for the models sensitive to the magnitudes of features.A number of machine learning models were trained and tested against each other such as Logistic Regression, Naive Bayes, Decision Tree, Random Forest, and Isolation Forest.

Model parameter settings were improved by implementing hyperparameter tuning through GridSearchCV and RandomizedSearchCV. To ensure that model selection is more robust and overfitting is less of a problem, tuning was done through cross-validation.Of the models built, Random Forest performed the best in terms of overall results and therefore was taken as the final model.

Some of the hyperparameters of the tuned Random Forest model include 100 trees, a maximum depth of 20, minimum samples split of 8, minimum samples leaf of 4, and a feature selection by a square root. The final model was assessed on the test set in terms of accuracy, precision recall, F1-score, ROC-AUC, confusion matrix, and feature importance analysis.

The experiments were conducted in Jupyter Notebook on a personal computer. The methods utilized in this work were traditional machine learning methods so there was no need for such things as GPU training, batch size setting, learning rate schedule, data augmentation, curriculum learning, or multi-stage training.

IV. Experimental Setup

A. Datasets

The project aimed at predicting diabetes with the help of health-related survey data from the BRFSS 2015 Diabetes Health Indicators dataset. This dataset includes demographic, lifestyle, and medical indicators such as BMI age hypertension, overall health, mental health income education, and mobility issues.

The output variable is Diabetes_binary, where 1 indicates a diabetic state and 0 a non-diabetic state. The raw material for the research was a dataset consisting of 253,680 samples and the task presented as binary classification with 2 classes. In the preprocessing phase, to prepare the data for training, the column names were cleaned, inputs and output were separated, and features were standardized. The dataset was split into training and test sets with 80% data for training and 20% data for testing. This amounted to 202,944 training instances and 50,736 test instances.

No separate validation set was formed since model tuning was conducted through cross-validation on training data.Among various diabetes datasets, BRFSS data is one of the best choices for this project as it includes real health-related factors associated with the risk of developing diabetes. The machine learning models can leverage the information from multiple health and lifestyle factors to learn the patterns and make predictions for diabetes diagnosis.

TABLE I. ***Summary of Dataset***

Dataset	Train	Test	Classes
BRFSS Dataset	202,944	50,736	2

B. Evaluation Metrics

Several evaluation metrics were used in this project to measure the performance of the machine learning models for diabetes prediction. Since the task is a binary classification problem, the selected metrics help evaluate not only overall correctness but also how well the model identifies positive diabetes cases.

Accuracy was used to measure the overall proportion of correctly classified samples. It is calculated as the number of correct predictions divided by the total number of predictions. Accuracy gives a general idea of model performance, but it may be misleading when the dataset is imbalanced.

Precision was used to measure how many of the samples predicted as diabetic were actually diabetic. A high precision means that the model makes fewer false positive predictions.

Recall was used to measure how many of the actual diabetic cases were correctly identified by the model. This metric is especially important in diabetes prediction because missing a true positive case can be more serious than making a false positive prediction.

F1-score was used as a balanced measure of precision and recall. It is the harmonic mean of precision and recall, making it useful when the dataset contains class imbalance and when both false positives and false negatives matter.

ROC-AUC was also used to evaluate the classification models. The ROC curve shows the trade-off between true positive rate and false positive rate at different decision thresholds, while the Area Under the Curve (AUC) summarizes this performance into a single value. A higher AUC indicates better ability of the model to separate diabetic and non-diabetic cases.

In addition, the confusion matrix was used to provide a detailed view of model predictions. It shows the number of true positives, true negatives, false positives, and false negatives, which helps in understanding the strengths and weaknesses of the model.

These measures have been selected as the dataset is essentially a binary classification umbrella that can harbor substantial class imbalance situations. Under such circumstances, leanness on accuracy alone can be a mistake. Precision recall F1-score, ROC-AUC, and confusion matrix collectively entail a more wholesome and dependable appraisal of the model performance.

C. Implementation Details

The project was implemented in Python using Jupyter Notebook. The main libraries used were pandas and NumPy for data handling, scikit-learn for machine learning models and hyperparameter tuning, and Matplotlib for visualization.

Since this project uses traditional machine learning models instead of deep learning, parameters such as learning rate, weight decay, batch size, optimizer type, and number of training epochs do not apply. These settings are commonly used in neural networks, but the models in this work were trained directly using the algorithms provided by scikit-learn.

The dataset was split into training and test sets using an 80:20 ratio, and feature scaling was applied before model training. Multiple models were tested, including Logistic Regression, Naive Bayes, Decision Tree, Random Forest, and Isolation Forest. Hyperparameter tuning was performed using GridSearchCV and RandomizedSearchCV with cross-validation.

The final selected model was Random Forest, which achieved the best performance. Its tuned hyperparameters were: 100 decision trees, maximum depth of 20, minimum samples split of 8, minimum samples leaf of 4, and square-root feature selection.

All experiments were run on a personal computer in a Jupyter Notebook environment. No GPU, distributed training, or special hardware acceleration was required for this project.

V. Results

A. Main Results

Table II presents a summary of the machine learning models' effectiveness in this research. The findings indicate that Logistic Regression and Random Forest were the top performers in accuracy, both achieving **86%**.

Furthermore, Logistic Regression was able to identify true positives with the least number of false positives (52% precision), unlike other models.In contrast, Naive Bayes led in recall (56%) and F1-score (41%) which indicates that it was more successful in detecting true cases of diabetes and had a better harmony between precision and recall. Its overall accuracy was not as high as that of Logistic Regression and Random Forest, but it was more effective in the positive class, which is a critical aspect in a medical prediction problem.Decision Tree yielded fairly good results while in terms of recall and F1-score, Isolation Forest was the most inefficient.

This is understandable as Isolation Forest is more oriented towards anomaly detection rather than typical supervised classification.In general, the comparison indicates that a single model is not superior in performing every metric.

Logistic Regression and Random Forest produced the highest accuracy whereas Naive Bayes excelled in recognizing positive diabetes cases. It is therefore a good practice to use various evaluation metrics instead of candling one's hopes on accuracy alone.

TABLE II. ***Comparison of Machine Learning models for Diabetes Prediction***

Method	Acc. (%)	Precision (%)	Recall (%)	F1 (%)
Logistic Regression	**86.00**	**52.00**	**16.00**	**24.00**
Random Forest	**86.00**	**49.00**	**18.00**	**26.00**
Naive Bayes	**77.00**	**32.00**	**56.00**	**41.00**
Decision Tree	**80.00**	**30.00**	**33.00**	**31.00**
Isolation Forest	**84.00**	**31.00**	**11.00**	**16.00**

The final model chosen for this project was the **Random Forest** classifier. Initially, the parameters were manually set and then hyperparameter tuning was done by GridSearchCV. As a result, the best parameter setting was n_estimators = 100, max_depth = 20, min_samples_split = 8, min_samples_leaf = 4, and max_features = sqrt.

With these tuned parameters, the model gave a test accuracy of **86.52%**. The ROC curve made by the last Random Forest model gave **AUC score 0.8211**, which is a good sign that the model can distinguish diabetic cases and non-diabetic ones. The confusion matrix tells us that the model correctly identified quite a few negative cases. However, it missed many positive diabetes cases. This is the same as the low class-specific recall for the positive class.

The classification report of the tuned Random Forest model indicated **precision = 0.57, recall = 0.14, and F1-score = 0.22** for the positive diabetes class. These results imply that even though the model gives good results on average, it is still favouring the majority class. This is a typical problem in highly unbalanced medical datasets. Feature importance analysis revealed that BMI, general health, high blood pressure, and age were some of the most important features. It means that these health-related variables had the greatest impact on diabetes prediction in our final model.

TABLE III. ***Final Tuned Random Forest Results***

Metric / Parameter	Value
Model	**Random Forest**
n_estimators	**100**
max_depth	**20**
min_samples_split	**8**
min_samples_leaf	**4**
max_features	**sqrt**
Test Accuracy (%)	**86.52**
AUC	**0.8211**
Precision (Positive Class)	**0.57**
Recall (Positive Class)	**0.14**
F1-score (Positive Class)	**0.22**

B. Ablation Study

An ablation experiment was done to gain insights on how different design decisions impacted the overall effectiveness of the diabetes prediction system. Since this work is reliant on conventional machine learning instead of deep learning, the ablation study reveals changes in model choice, hyperparameter tuning, and feature contribution rather than removal of layers of neural networks.

At first, various types of machine learning models were experimented with, such as Logistic Regression, Naive Bayes, Decision Tree, Random Forest, and Isolation Forest. According to this evaluation, the choice of the model strongly determined the performance. Logistic Regression and Random Forest could achieve the highest precision, whereas Naive Bayes led to the highest recall and F1-score. This tells us that the model is a crucial design element of the project.

The second point is that the hyperparameter tuning was considered for the Random Forest model. The adjusted Random Forest model with parameters such as **n_estimators=100, max_depth=20, min_samples_split=8, min_samples_leaf=4, and max_features=sqrt** gave better results than a default or less set scenario. Hence, it is proved that the performance of the final model was uplifted by the optimal selection of parameters.

The third is that the feature importance plot was the means for knowing the role of the independent variables. One of the outcomes of this analysis is that BMI, general health, high blood pressure, and age are the four major predictors. This leads to an interpretation that the model still mainly depends on these highly relevant features and the less important ones contribute less to the overall prediction.

To sum up, the ablation experiment reveals that the ultimate model is a combination of three major design decisions: opting for Random Forest as the main model, tuning its hyperparameters, and selecting health-related features as informative. The results support the structure of our proposed diabetes prediction pipeline.

C. Qualitative Analysis

Fig. 2. displays the confusion matrix and feature importance results of the ultimate Random Forest model. The confusion matrix illustrates that the model was able to correctly identify a high number of non-diabetic cases however it overlooked a considerable number of diabetic cases. In other words, the model was successful with the majority class but was unable to detect all the positive cases. On the other hand, the feature importance diagram indicates that BMI, general health, high blood pressure, and age were the top features contributing to the prediction. These charts provide the rationale behind the advantages and disadvantages of the final model.

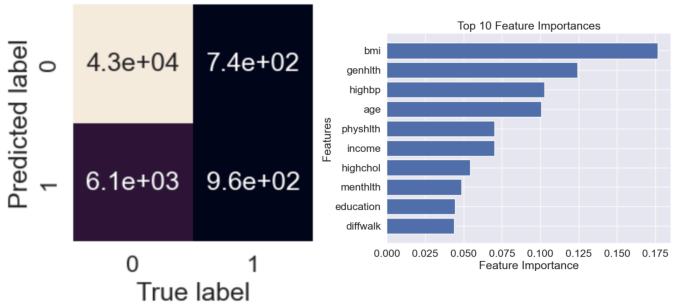

Fig. 2. Confusion Matrix and Feature Importance chart

VI. Discussion

This explains one way machine learning algorithms can enhance diabetes diagnosis, namely through diverse health indicators. The most accurate Random Forest model recognized meaningful patterns in the data very well as evidenced by its high overall accuracy. Additionally, the discovery of feature importance led to the identification of BMI, overall health, hypertension, and age as the leading factors in making predictions.

This, in turn, made the model more interpretable. However, the results also bring out some limitations.The model showed a pretty good overall accuracy but its recall for the positive diabetic class was quite low. As a result, a large part of actual diabetic cases was overlooked. One of the major reasons for that is probably the imbalance of classes within the dataset when non-diabetics are much more frequent than diabetics. So, the model often leans toward the dominant class.

Besides, the project relies on just one public dataset and, as a consequence, its findings might not be extendable to other populations or clinical environments. Also, it is possible that certain key medical or lifestyle factors are insufficiently represented in the data. The model might also perform badly in those cases where a patient's health condition is fairly unusual or does not match typical patterns in the training data.

Overall, the methodology produces useful results.Nevertheless, upgrading the class imbalance handling, testing on different datasets, and using techniques that bring about positive diabetes case detection enhancement are some of the points that can still be considered.

VII. Conclusion

Our project proposed a machine learning method for predicting diabetes utilizing the BRFSS 2015 Diabetes Health Indicators dataset. Several models such as Logistic Regression, Naive Bayes, Decision Tree, Random Forest, and Isolation Forest were trained and their performance was compared. It was evident from the experiments that the choice of model significantly influences the performance and a single model is not necessarily the best for every metric.

Of all the models, the Random Forest was finally chosen as it not only performed well but also showed a clear feature importance analysis. The research further identified BMI, general health, high blood pressure, and age as the primary factors for prediction. However, the results also reflect the issue of class imbalance since the model failed to identify many positive diabetes cases.

In general, this project demonstrates that machine learning can assist in diabetes prediction based on structured health data. Moving forward, the work can be extended towards better handling class imbalance, experimenting with other datasets, and exploring more sophisticated models or feature selection methods.

Acknowledgement

Our first thought was to handle the project with unsupervised clustering techniques like K-Means and Fuzzy C-Means. On the other hand, our deep exploration of the CDC Diabetes Health Indicators dataset revealed that it is targeted for a binary classification task and is characterized by a significant class imbalance. We had a conversation with our professor, Dr. Tanvi Banerjee, about it and she pointed out that we should not try to fit an unsupervised method forcibly but look into the anomaly detection techniques like Isolation Forest and then compare them with the traditional machine learning models. Consequently, we gave our project a new shape following her advice, and this led to our current version of the research.

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Author Contributions

Prapti Acharya: Conceptualization, Software, Data Collection and Curation, Model development without hyperparameter tuning (Logistic Regression, Naive Bayes, Decision Tree), Model Evaluation and Performance Analysis, Visualization, Final Report Writing, Final Report Compilation.

Kiran Bhandari: Methodology Design, Random Forest Model Development, Hyperparameter Tuning (GridSearchCV, RandomizedSearchCV), Isolation Forest Implementation, Validation, Initial Report Writing, Project Supervision and Final Report Compilation.